JMO – 2018

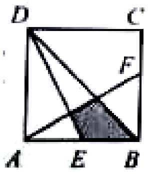
Answer all questions. All questions carry equal marks.

Full Marks 100

1. Two oxen are sold at Rs. 8000 each. If a gain of 20% is obtained in one and a loss of 20% is incurred in the other, then determine the total percentage of gain or loss.
2. 8 persons complete 8 pieces of work working for 8 days at the rate of 8 hours per day. If 4 persons work for 4 days at the rate of 4 hours a day, how many pieces of work can be completed?
3. Determine the greater number: 39^{40} *or* 40^{39}
4. Each distinct letter represents a digit. Find the number represented by the sum in the following addition:

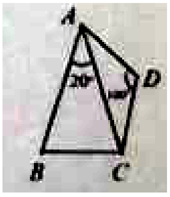
$$\begin{array}{rr} & \text{WHITE} \\ + & \text{WATER} \\ \hline & \text{PICNIC} \end{array}$$

5. Let a, b, c be real numbers such that $a^2 - 2 = 3b - c,\ b^2 + 4 = 3c + a,\ c^2 + 4 = 3a - b$. Find $a^4 + b^4 + c^4$.
6. Let r be a real number such that $\sqrt[3]{r} - \frac{1}{\sqrt[3]{r}} = 2$. Find $r^3 - \frac{1}{r^3}$.
7. In an answer sheet consisting of 16 pages numbered 1 to 16 some pages are torn. If the sum of the page number of the remaining pages is 86, determine the page numbers of the torn pages.
8. Determine the number which when multiplied by 999 the product will have all its digits 1.
9. Determine the digits of the tenth place and unit place of 3^{2018}.
10. Smaller angle between the minute hand and the hour hand at 10 a.m is 60^0. At what time next the smaller angle between them will be 60^0 again.
11. Chandan was asked to find the LCm of 27 and 45 and another number. But he computed the LCM replacing 27 by 72 and other two numbers correctly. If the LCM of the given numbers and the LCM computed by Chandan are same, then determine the least value of the third number.
12. How many four digits numbers are there having exactly three equal digits.
13. In the square ABCD, E and F are mid points of AB and BC respectively. Determine the area of the shaded portion as a fraction of area of ABCD.

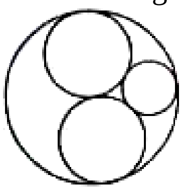


14. Let a, b, c be three nonzero real numbers and $a + \frac{1}{b} = 5,\ b + \frac{1}{c} = 12,\ c + \frac{1}{a} = 13$. Find the value of $abc + \frac{1}{abc}$.
15. For alcohol in water solution how many ml of 15% solution be mixed with a 35% solution to get 250 ml of 21% solution.
16. If m, n be positive co-prime integers and $\frac{1}{1+2} + \frac{1}{1+2+3} + \frac{1}{1+2+3+4} + ... + \frac{1}{1+2+3+...+20} = \frac{m}{n}$, find $m + n$.

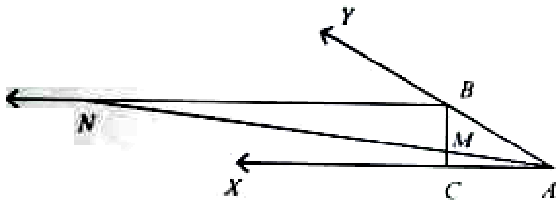
17. Solve system of equations: $xy + x + y = -13,\ yz + y + z = -9,\ zx + z + x = 5$, where x, y, z are integers.
18. In a market complex the shops are numbered consecutively from 1 to N and total 3189 digits are used in all. How many shops are there in that market complex?
19. The highest score in mathematics test was 100 and lowest score was 81. At least 3 students in the class had the same score. What could be the minimum number of students in that class (Assume all scores are whole numbers).
20. In a quadrilateral ABCD, AB = AC = 15 cm, AD = CD = 8 cm, $\angle BAC = 20^0$, $\angle ADC = 100^0$. Find BC.



21. A circle of radius 24 cm touches two circles of radius 12 cm each internally and two small circles of radius *r* touch each other externally as shown in the diagram. Find *r*.



22. In the given figure BC is perpendicular to AX. BN is parallel to AX and MN = 2AB. Prove that $\angle YAX = 3\angle XAN$.



23. In the given figure cross made up of five congruent squares is inscribed in a circle. If the length of side of each square is 14 cm, then determine the area of the shaded region of the figure.

******
****